Second Edition

RESEARCH METHODOLOGY IN APPLIED ECONOMICS

Organizing, Planning, and Conducting Economic Research

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Methodological Concepts and Perspectives

After a certain high level of technical skill is achieved, science and art tend to coalesce in esthetics, plasticity, and form. The greatest scientists are always artists as well.

Albert Einstein

Research methodology in economics is obviously not independent of the perspectives and boundaries of the discipline. It also conforms to certain long-established rules of the scientific and academic community. This chapter defines, discusses, and explains some general aspects of research methodology and its relationship to science, economics as a social science, theory, facts, objectivity, and such matters. The chapter begins by defining science, then progresses to considering knowledge, including classification of types, the role of personal objectivity, and the roles of facts, theory, and hypotheses.

SCIENCE

We may note that the term *science* has become associated with only the physical and biological sciences (chemistry, biology, geology, etc.) during the most recent century or so. McCloskey (1990b, p. 7) explains that "other languages use the science word to mean simply 'systematic inquiry.' . . . Only English, and only the English of the past century, has made physical and biological science into . . . 'the dominant sense in ordinary use.' " The treatment of science in this book is as a systematic inquiry rather than as a predesignated set of disciplinary areas.

Research is designed to generate new, reliable knowledge. The primary vehicle for achieving that reliable knowledge is science.

> Science is the organized accumulation of systematic [reliable] knowledge for the purpose of intelligent explanation/prediction. (Williams, 1984)

The accumulated knowledge, the result of the processes of discovery and confirmation, is typically grouped into categories or branches of learning or instruction (e.g., chemistry, engineering, economics, physics, art, history), which we identify as disciplines or areas of study.

Several aspects of this definition are important. Note that the intended purpose of science is explanation/prediction. The slash is deliberately used because in the meaning of science, the two terms are intricately intertwined. By explaining what occurs and how or why it occurs, we develop the capacity for *conditional* prediction. That is, we are able to specify that "if X, Y, and Z occur, then W will follow." Scientific prediction is always conditional prediction. Unconditional prediction (e.g., foretelling the future) must have some means of knowing what conditions will be in effect (knowing that X, Y, and Z will, in fact, occur), and at least some of those elements are beyond the control or ability of the scientist to forecast with certainty. Thus, even when the conditions of a forecast are not made explicit, the predictions are still conditional.

While explanation is the primary goal, knowledge is the means of science. Science gives us the tools (accumulated knowledge) with which to do research, and research adds to the accumulated body of reliable knowledge. The two are mutually interdependent. Yet they are different, and the distinction is important. Science (or science*s*) as a body of accumulated reliable knowledge is an entity; but it is not a static or unchanging entity. Research is a process through which (1) the body of science is expanded; and (2) the existing body of science is tested for correctness or validity. The two are symbiotic—research draws on the body of accumulated knowledge as an integral part of its process, and the accumulated knowledge is expanded through the research process.

Research methodology is the methodology of science in the sense that it is the methodology of *expanding* science. Research is the means of accumulating, evaluating, and questioning bodies of knowledge we call science, as well as a process of using science in addressing issues, problems, and questions. Research and science are intricately inseparable. We distinguish between them because both are important separately, but neither can function without the other. As individuals, we cannot practice research without a grasp of existing disciplinary, subject-matter, and problem-solving knowledge (science), and we have no means to expand that knowledge except through a process of creating new reliable knowledge or understanding (research).

On the matter of science, we may as well dispel a popular misconception that "science" consists only of factual "truth," devoid of any human values and unaffected by private views. Simkin (1993, p. 12) states: "Metaphysics (i.e., philosophy or beliefs) and science have, of course, been closely related, both historically and logically." The Committee on Science, Engineering, and Public Policy (1995) states that science is inherently a social enterprise (p. 3), and that scientific knowledge emerges from a process shaped by human values, limitations, and social contexts (p. 2). The committee further notes that personal beliefs (philosophical, religious, cultural, political, and economic) can affect scientific judgment, but when judgment is recognized as a scientific tool, it is easy to see how science can be influenced by values (p. 6). Additionally, these social mechanisms and values both validate scientific knowledge and sustain methods and conventions for doing research (p. 4). This does not diminish the importance of science or its role in our pursuit of knowledge; it merely reminds us that scientific pursuits are carried out within social and individual systems of values, and by people who are fallible. We must recognize the role of individual, social, and scientific values in scientific pursuits and be diligent against blind acceptance (or rejection) of ideas, observations, or concepts. Put another way, science teaches both rational thought and freedom of thought (Feynman, 1999, p. 186).

Before progressing further into methodological matters, we will digress briefly into a discussion regarding economics as a science. While the views presented on the matter are not unique, they may serve to encapsulate the issues of a long-running and ongoing debate between economists and others—a debate that, although it will almost certainly continue, does not warrant substantial consideration.

ECONOMICS AS ART AND SCIENCE

To better understand this issue, some classifications of science are useful. Note that these are classifications, not definitions. We can break the sciences into the physical sciences (e.g., chemistry, biology, physics) and the social sciences (economics, sociology, anthropology, etc.), the distinction being that the physical sciences are concerned with things while the social sciences are concerned with people (Stewart, 1979, p. 5). The physical sciences are sometimes called the "hard" sciences. However, Johnson (1986b, p. 31) notes that they are really the "easy" sciences because they deal with inanimate things rather than human social behavior. Nevertheless, these groups, and the difference in focus between them, may be a key to understanding the debate.

The physical sciences were the first to adopt the designation of *science*, and they developed and evolved with a heavy orientation (belief?) in the philosophy of positivism (to be covered in chapter 4) and a strong empirical tradition—a reliance on and preoccupation with observations and data. Furthermore, in their developmental stages, several sciences evolved in a period and under a set of conditions in which the only feasible way to hold external forces constant was in a laboratory setting. Consequently, the laboratory, and the classical laboratory experiment, became their sole or primary "model" for being designated as "scientific."

The social sciences, on the other hand, study human behavior and phenomena that do not lend themselves to controlled or laboratory experiments. Controlled experiments in the social sciences are limited either because society does not find such experiments ethically acceptable or finds them prohibitively expensive. In place of laboratory-generated data, social scientists, and especially economists, have substituted and developed more sophisticated multivariate conceptual and statistical techniques to adjust for a myriad of forces that affect the major variables of interest. Further, economics is influenced by normativistic philosophy (also covered in chapter 4), which accepts social and individual values as being entities open to scientific inquiry. Positivists would accept the study of things as "real" because they have observable, physical dimensions, but values cannot be "real" because they have to do with states of mind. Those who exclude economics as a science because it fails to use a traditional laboratory or because of its range of subjects of study are defining science in terms of specific *methods* rather than in terms of the *methodology* (the general approach).

Another distinction between physical and social sciences lies in the relative emphasis placed on the balance between theory and data. While all sciences use both, a valid generalization is that physical sciences place more emphasis on data, while the social sciences place more on theory (these concepts will be defined more precisely and discussed further later in this chapter). Incidentally, these distinctions are not independent of the differences discussed above. The physical sciences generate data under controlled conditions as a means of testing their theories. These theories are, in simple terms, postulated relationships between inanimate forces, given specified conditions or other factors. As such, these sciences have a strong empirical tradition of verifying their theories through experimental design and data generation. The theories of the social sciences tend to address more complex phenomena of individual and group motivation and behavior and the effects of societal institutions on them. These phenomena are often not capable of being directly observed or quantified under artificially controlled conditions (Leontief, 1993 provides an insightful discussion of this issue). This does not, however, necessarily make them less real. On the other hand, some economists (e.g., Eichner, 1986) argue that economics has failed to confirm its theories empirically to the degree required by "science."

Having presented this view, we may also recognize that these generalizations weaken under extended examination. Consider, for example, that physics as a physical science relies more heavily on conceptualizations and somewhat less on controlled experiments than some other physical sciences. This is largely because many phenomena studied are complex, and controlled data are difficult. With regard to economics, Johnson (1986b, p. 82) notes that the criticism that economists are lax in using data to validate or test theory is directed more at the disciplinary work of economists than at the subject-matter and problem-solving work within the various fields of the discipline. McCloskey (1983, p. 491) states:

> Economics is not a science in the way we came to understand that word in high school. But neither, really, are other sciences. . . . Economics is misunderstood and . . . disliked by both humanists and [physical] scientists. The humanists dislike it for its baggage of antihumanist methodology. The scientists dislike it because it does not in reality attain the rigor that its methodology *claims* to achieve.

McCloskey (1990a, p. 1127) also argues that economics is not sufficiently concerned with the empirical validation and testing (empiricism) to rank with the physical sciences, and that more focus on empirical methods in applied research would make it scientifically stronger.

Whether economics is an art or a science constitutes a stimulating philosophical topic for debate, but one's position depends on the initial premises one adopts. Economics is, like many disciplines, *both* art and science. It is a science in terms of its accumulated knowledge base, its descriptive knowledge, and its procedures. This includes economic research activities. It is both a science and an art in terms of applying that knowledge to current issues and problems. This includes both research and nonresearch (decision and action) types of activities. As economists, we know that the government setting a price floor on a good that is above the market equilibrium will cause a surplus of that good to accumulate, ceteris paribus. Yet the answer to the question, "What will happen if the price support for wheat in the United States is increased by 10%?" depends on how well we can anticipate what will happen to a host of other variables. These external variables may include levels of wheat production and income in other countries in the world, whether U.S. policy makers change other programs (e.g., food aid, export enhancement programs, acreage controls), and whether weather patterns in U.S. wheat-production areas are normal. However, in the application of science, the chemist also must anticipate effects of uncontrolled factors when predicting chemical reactions outside the laboratory. Thus, we all practice science to some degree within some boundaries, but include the practice of art, and become more multidisciplinary, when we attempt to make it applicable to problems and issues that are of concern in the larger "practical" world in which we participate. Research encompasses both art and science.

KNOWLEDGE

In considering reliability of knowledge, one of the problems is lack of agreement on what constitutes "evidence," but it is the reliance on evidence that constrains the influence of personal values in science (Committee on Science, Engineering, and Public Policy, 1995, p. 8). McCloskey (1983, p. 491) argues against constraining evidence to an artificially narrow range of criteria (e.g., the evidence must be "objective" or "experimental" or "observable"). In our examination of knowledge and its reliability, we will consider two classifications of knowledge, both of which are useful for purposes of examining research methodology. One classification system is by Larrabee (1964) and the other is by Johnson (1986b).

Positivistic vs. Normativistic Knowledge

The classification system by Johnson (1986b, pp. 16–20) specifies knowledge as positivistic knowledge and normativistic knowledge, with normativistic being subdivided into two types: prescriptive knowledge and knowledge of values. *Positivistic knowledge* is knowledge of conditions, situations, or things that are directly observable or measurable. *Normativistic knowledge* is knowledge about values. *Prescriptive knowledge* is knowledge of what ought/ought not be done to solve a specific problem. *Knowledge of values* is knowledge of the goodness and badness of conditions, situations, and things. The effects of the conditions, situations, and things are observable, but the inherent goodness or badness of the effects is a matter of perspective and interpretation. Knowledge of values includes knowing how people evaluate conditions, situations, or things in terms of good and bad, whether on some quantified scale or as binary entities. It can also be perceived as knowledge of what has value even though no one realizes it. When we determine, individually or collectively, that conditions, situations, or things are inherently good or bad, that knowledge is subjective. But knowledge of values is not inherently subjective; it may be the result of an evaluative, analytical, or measurement process and, as such, be as objective as positivistic knowledge.

Prescriptive knowledge is concerned with decisions about good and bad in a particular situation and inherently embodies judgment; it may be a logical extension of a combination of both value-free knowledge and knowledge of values. Prescriptive knowledge is necessary for individuals, groups, and society at large to make decisions and take actions. We all possess prescriptive knowledge, but it is conditioned by our own states of mind, personal and collective biases, experiences, perceptions of the world, moral/ethical conditioning, and other such factors.

We can make an additional differentiation within prescriptive knowledge. Prescriptive knowledge can be unconditional or conditional. *Unconditionally prescriptive knowledge* takes the form of "Prescription X is best (irrespective of the circumstances)." Persons claiming unconditionally prescriptive knowledge assume the attitude that their prescriptions are above question. Conditionally prescriptive knowledge is tempered by circumstances. It takes the form of "If the objective is Y, then the most effective approach is prescription X." *Conditionally prescriptive knowledge* recognizes that prescriptions are conditional on objectives and constraints.

Of these classes of knowledge, unconditionally prescriptive knowledge is handled cautiously in the research process. Recall that the goal of the research process is *reliable* knowledge, or that which is acceptable based on evidence. Reaching agreement on unconditionally prescriptive knowledge based on evidence is usually a severe task. This does not say that unconditionally prescriptive knowledge is "bad." All social entities must reach some agreements on unconditionally prescriptive knowledge (e.g., that first-degree murder requires a defined minimum penalty). However, those agreements are not reached through research. Research that deals with, for example, the effects of given actions, may contribute to those agreements, however. For purposes of our consideration of research methodology, references to prescriptions will be understood to be conditional unless otherwise noted.

It should be noted that there is no clear consensus among economists on the relative importance of conditionally prescriptive knowledge in the research process itself. For example, Edwards (1990, p. 5) maintains that prescription is *the* reason for doing research. Those who hold this view are focused primarily on policy or other decision or action applications of research. Those who confine themselves to disciplinary research may view prescriptive knowledge as less important. The differences in views about the role of prescription in research center on where to draw the line between research and decisions, particularly in cases when research is conducted with policy application in mind. This distinction is considered again later in this chapter and in chapter 6.

Private vs. Public Knowledge

Another way of classifying knowledge is as private or public knowledge (Larrabee, 1964).[1] *Public knowledge* is knowledge that can be demonstrated to others through logic or evidence. Public knowledge is, by definition, reliable knowledge. Reliable knowledge is public in nature; it is not depleted when shared (and often is expanded by sharing), and others cannot be excluded from its use (Stephan, 1996, p. 1200). *Private knowledge* is that which we accept or "know" for ourselves but that cannot be demonstrated to others. We all "know" (i.e., feel, believe, trust, etc.) things that we accept as reliable for ourselves but have no way to demonstrate to others. Perhaps the clearest example of private knowledge lies in the area of religious beliefs. Many of our individual or collective religious beliefs constitute private knowledge because there is no means to demonstrate the validity of the knowledge using logic or evidence. Acceptance of its validity rests on "faith."[2] It should be recognized, however, that knowledge in any field or discipline is private knowledge if its reliability cannot be accepted on the basis of evidence. Consider, for example, the knowledge that "capitalism is better than socialism." The reliability of this subjective proposition cannot be demonstrated because the criteria depend on one's view of what constitutes "better." On the other hand, the knowledge that "capitalism with competitive markets allocates goods and services more efficiently than does socialism" can be demonstrated by logic and evidence, given a specific definition of efficiency.[3] These distinctions will be further discussed in terms of subjective and objective statements and states of mind.

Private knowledge is used with caution in the realm of research. Note that private knowledge is not necessarily ranked as inferior, but we have no way to employ it to reach conclusions through research. Only public knowledge can be reliable in the scientific sense. However, private knowledge may lead to public knowledge when the circumstances allow it. When an individual intuitively understands something (possesses private knowledge) and sets about demonstrating the reliability of it, it may achieve the status of public knowledge. Much public knowledge likely started as private knowledge, and public knowledge (what is accepted on the basis of logic and evidence) is culturally and time dependent. Many of us have experienced this drive to demonstrate, show, or "prove" some aspect of our private knowledge to another person, the public, or the scientific community. We should recognize, however, that the desire to turn private knowledge into public knowledge has at least two major pitfalls: the potential incursion of subjectivity and the potential for the logical fallacy of special pleading. When we want something to be demonstrated because we have a vested interest in it, we are less likely to examine the evidence against it as carefully as the evidence for it and may, therefore, be more likely to present only the arguments that favor its acceptance.

WAYS WE OBTAIN KNOWLEDGE

We gain knowledge through six primary means: the senses, experience, intuition, revelation, measurement, and reasoning. While this list is not exhaustive, and they are not fully independent of one another, all avenues to knowledge can be placed under one of these categories, or some combination of them. The ensuing discussion attempts to briefly explain each.

A. *The senses.* Knowledge gained through the senses—sight, sound, touch, taste, and smell—may be regarded as the first step toward gaining private knowledge. It is also, as noted previously, the principal means of deriving value-free positivistic knowledge. As infants, we began accumulating knowledge through our senses, and we continue to rely on them for gaining knowledge about people and things throughout our lives. We might even classify this knowledge as "private facts." The things we taste, see, touch, and so on provide us with factual information. This knowledge may remain private rather than public because (1) our perceptions through our senses differ among individuals; and (2) the sensory information may not be capable of being demonstrated: there is nothing certain in our sense experiences because we interpret in light of what we have already learned (Simkin, 1993, p. 23). On the other hand, this knowledge forms part of the basis for each of the other avenues to knowledge except reasoning.

B. *Experience.* Experience constitutes the accumulated total of one's exposure to and interaction with people, places, things, circumstances, ideas, senses, and so on. Knowledge gained through experience may or may not be reliable, depending on its nature. Experience knowledge is essentially private until measures are taken to make it public. Experience, being what it is, is made of very divergent elements—a hodgepodge of information—and creating new reliable knowledge from it is typically a cumbersome process. Stated another way, it is a disorderly and unorganized means of learning (Williams, 1984). It is, however, often an essential component of making sense of knowledge that we possess or accumulate; we use our experience to evaluate new knowledge in its relation to prior knowledge.

C. *Intuition.* Intuition is the sensing or feeling of something being accurate or not. If it constitutes "knowing," it is knowledge in a vague form. As such, it is inherently private knowledge only. However, an intuitive understanding of a fact, relationship, or set of relationships may lead to an orderly exploration and logical development, which leads, in turn, to public knowledge. Back (no date) suggests that intuition may be a necessary condition for successful research, but it is not a sufficient condition for finding reliable knowledge. Intuition is often trustworthy as private knowledge and may lead to public knowledge, but in research and science it cannot be accepted on its own as reliable. As noted in chapter 2, Ladd (1987) is a strong advocate for the role of intuition in the creative research process, but he makes no claims that it can establish reliable knowledge on its own.

D. *Revelation.* Revelation, or the reception of knowledge from a vaguely defined or unknown source, is somewhat disquieting from a purely scientific perspective. It is not confined to divine revelation in a religious context. Most of us have experienced an idea "out of the blue": something that just occurred to or dawned on us. This is a form of revelation. It may spring from the unconscious mind and it probably relates to intuition, perhaps experience. Revelation as a source of knowledge is, however, not reliable. It may become reliable knowledge only if it meets the tests for reliability.

E. *Measurement.* Knowledge gained through measurement (quantification) is normally thought of as data. Its connection to the senses and experience is obvious. This is usually regarded as factual knowledge. It is also regarded as reliable most of the time, with the implicit understanding that there may be a sampling or measurement error associated with it. Some disciplines devote a large share of scientists' time and attention to acquiring factual knowledge through measurement. As suggested in earlier discussion, economics does not usually emphasize laboratory or field measurement of data in a physical sense, but statistical sampling and gathering data with surveys are important parts of economic research activity.

F. *Reasoning.* Reasoning may be viewed as the "final test" of knowledge or gaining knowledge. Knowledge obtained through reasoning is the only way to derive reliable knowledge of relationships and patterns through which we develop explanatory or predictive capability. Of the two avenues to reliable knowledge—measurement and reasoning—reasoning constitutes the only avenue to useful disciplinary, subject-matter, and problem-solving knowledge. Reasoning, being deductive, inductive, or both, is the way we establish the relationships, patterns, concepts, and basic theories through which we can make facts or data mean something that has relevance—to disciplinarians, managers, policy makers, consumers, voters, and so forth.

This does not say that reasoning alone can establish reliable knowledge, only that if knowledge cannot pass as logical it is not trustworthy. Ladd (1991, p. 4) points out that it is not possible "to prove anything about the real world by deduction because it is not possible to prove deductively that real types have the properties attributed to ideal types." Induction is fallible because hypotheses are derived from several assumptions, and observations that contradict a hypothesis indicate that at least one assumption is wrong but do not identify the wrong ones (p. 5). Consequently, both deduction and induction are fallible as tests of final reliability. Yet reliability cannot be established without one of them, often both.

RELIABILITY OF (PUBLIC) KNOWLEDGE

In our examination of knowledge and its scientific reliability, the question of criteria for reliability is critical to the notion of validity in a science or discipline (being both correct and meaningful). The criteria are confined to public knowledge. The criteria are also somewhat nebulous because they ultimately reduce to something on which reasonable people can agree. There are two general criteria for establishing reliability of knowledge: (1) it can be supported by evidence; and (2) the way the evidence is obtained or generated can be demonstrated or reproduced. This evidence may be quantitative (i.e., data) or it may be more complex logical constructions that include data but proceed on to relationships, generalizations, or deductions/inductions from the data. In the first instance, the observed phenomena may be measurable, or at least capable of being described or represented (e.g., photographed, drawn). We think of this as factual information or knowledge. But how do we establish the reliability of things that we cannot observe directly? We do it through reasoning, or logic.

Logic may be classified as being of two general types, deductive and inductive.[4] *Deductive logic* is the process of reasoning from general premises (e.g., assumptions) to specific results or conclusions. Economic theory rests largely on deductive logic. We establish a set or series of assumptions about conditions, motivations, and behavior, then logically work through to the expected outcome in terms of some set of variables or parameters that we wish to explain or predict. Theories of market behavior start with assumptions about market structure, people's motivations and objectives, and influence of external forces, then deduce a set of behaviors of market prices and quantities from those conditions. Mathematics may be the most clear-cut example of deductive logic. It begins with a set of definitions (of numbers, sets, etc.) and derives, through deduction, a large, complex system of theorems, laws, rules, and so on about behavior of numbers and systems of numbers.

Inductive logic is reasoning from the specific circumstances or outcomes to a conclusion about general circumstances or outcomes. It may also be represented as an empirical process of reaching a conclusion or arriving at new principles from known data and experience by observing objective realities (Ghebremedhin and Tweeten, 1994, p. 10). The most applicable explanation of induction for economics is statistical inference. By structuring a random sample of a larger population, collecting data about that sample, and analyzing the data using established criteria and procedures, we can infer characteristics and behavior about the entire population from the sample analyzed. Induction is used extensively in economics, especially in empirical research but in other ways as well. Additionally, the conclusions drawn from economic research—the generalizations about the larger world based on analysis or study of portions of it—are based on inductive reasoning.

LOGICAL FALLACIES

Whether the logic or reasoning is inductive or deductive, there are some logical fallacies, an awareness of which can make scientific inquiry (i.e., research) more efficient. We may note that to identify something as a fallacy does not say that it is automatically not so, but that it is not *necessarily* so. The description of what constitutes logical fallacies may vary with one's focus and intent. Different authors list them differently, some with Latin names, others with more descriptive terms (see, e.g, Stewart, 1979, pp. 20–29; Spencer, 1971, pp. 9–11).

The following list of logical fallacies is reasonably complete for our purposes.

A. *Special pleading*. This logical fallacy occurs when one uses only the information that supports a predetermined position or conclusion, while ignoring other information. This logical fallacy is common in law and politics, even accepted as a desirable approach in trial law. Its pitfalls in the pursuit of reliable information or knowledge are obvious.

B. *Affirming the consequent*. This logical fallacy occurs when conclusions are based on premises without examining the validity of the premises. An example is the unqualified conclusion that "free markets" always lead to an optimum social welfare. We know that optimum social welfare can occur under certain circumstances, but the existence of monopolistic power, market externalities, and other factors can negate that result.

C. *Attacking the person*. This fallacy involves rejecting a position or conclusion because of attitudes about the person or group presenting the position or conclusion. It may take the form of "What individual X says is false because X is a Democrat (or Republican or black or Catholic, etc.)."

D. *Appeal to the people*. This involves accepting or rejecting a position or conclusion because a large number of people accept or reject it. This fallacy has some elements in common with the fallacy of affirming the consequent.

E. *Appeal to authority*. This fallacy occurs by automatically accepting a position, proposition, or conclusion because its source has specialized or in-depth knowledge of it. It may take the form of "Statement A is true because individual (or group) X says it is true." This fallacy also has some obvious common elements with affirming the consequent and can be viewed as a variation on the fallacy of appeal to the people.

F. *False cause*. This fallacy (also called the post hoc fallacy) is the danger of attributing the wrong cause to an effect. It reasons that "Because event B occurs after event A, A causes B." We are particularly susceptible to this logical fallacy from spurious correlation (when things occur together by sheer happenstance—there is no possible connection) or when another set of conditions causes both to occur. An illustration: California has the highest per capita consumption of wine in the United States and the highest incidence of blindness: therefore, wine causes blindness.

G. *Argument by analogy*. This fallacy reasons that "A is similar to B, so what is true for A is also true for B." A common application of this fallacious reasoning is that because certain government policies are effective, they also would be effective in China (or France or Zaire, etc.). Although by itself a logical fallacy, argument by analogy can also be a tool for discovery. That is, something that works in one situation may also work in another, but it cannot be *automatically* assumed to work. This fallacy also is related to affirming the consequent.

H. *Composition and division*. These are presented together because they are mirror images of one another. The fallacy of composition reasons that what is true for a part (an individual) is automatically true for the whole (the group). An economic example of this is "Since an individual consumer cannot affect the price of milk, all consumers collectively cannot affect the price of milk." The fallacy of division reasons that what is true for the whole is automatically true for each part. An example is that because a nation's income is greater when consumption increases, individuals' incomes are automatically greater when they consume more. These fallacies are well understood by economists because examples of them are explicit in economic theory.

TESTS FOR RELIABILITY

The above considerations apply to reasoning or logic alone. When we rely on other input such as data, observation, or empirical evidence (alone or in addition to logic) in the pursuit of reliable knowledge, avoiding logical fallacies is not sufficient. Furthermore, we usually use other input, particularly in applied research. Thus, for a more complete consideration of reliability, we need more comprehensive guidelines. When evaluating overall reliability, there are at least three tests that are recognized. These are the tests of logical coherence, correspondence, and clarity (sometimes called comprehensiveness) (Johnson, 1986b, pp. 43–49; Eichner, 1986, pp. 5, 6). We also can consider a fourth test—the pragmatic test of workability. Coherence is a logical test, constituting a necessary, but not sufficient, condition for reliability; the other tests are essentially empirical tests.

The *test of logical coherence* is examining an outcome or proposition to see if it is free of logical contradiction. This amounts to examining the outcome or proposition to see if it is free of logical errors, including the logical fallacies identified in the previous section. When empirical observation or data are involved, meeting the test of logical coherence does not ensure the reliability of the outcome, but its reliability fails if the test is not met. When applied to theory, it is a test of whether the conclusions follow logically from the assumptions.

The *test of correspondence* involves comparing an outcome or statement to what is already "known" to see if it is consistent with (corresponds to) prior knowledge. If a result is consistent with prior knowledge or results, it passes the test of correspondence. When applied to theory, it examines whether conclusions are supported by empirical evidence. Note that this does not say that we place unconditional trust in conventional wisdom. In fact, new findings that contradict what was previously "known" is a primary avenue to new knowledge. However, meeting the test of correspondence constitutes one piece of evidence to support a result, statement, or position.

The *test of clarity* examines the outcome or proposition for lack of ambiguity or vagueness. If the result or proposition has more than one meaning, it fails the test of clarity. Johnson (1986b) uses overidentified and underidentified reduced-form simultaneous model equations to illustrate a lack of clarity. Another illustration of lack of clarity is the development or application of a concept when the terms used in the concept are not defined, rendering their meanings ambiguous. When applied to theory, it is a test of whether the theory is consistent with all the facts pertaining to the theory (Eichner, 1986, p. 5). Unless a result or proposition meets the test of clarity, applying either of the other two tests may not be possible.

The pragmatist's *test of workability* (pragmatism will be addressed in more detail in chapter 4) requires that the result solve the problem or issue addressed. The outcome or proposition must achieve the desired result; that is, it must "work." The addition of this test as a test of reliability is relevant, of necessity, for economists engaged in problem-solving research, but not necessarily for those focused on disciplinary research. Thus, the view advocated here is that workability may be a useful test in given circumstances, but not as a test of reliability to be applied universally.

A simple example of these reliability tests may help to illustrate their uses.[5] Consider the situation of a physician giving a child a smallpox vaccination. Does the action meet the test of correspondence? For the physician, yes, because he or she knows that it has worked in all previous cases. For the child, probably no, because he or she lacks prior knowledge of the results. Does it meet the test of coherence? It does to the physician (who understands the concept) but not to the child (who does not understand how being hurt can help). The action meets the clarity test for the physician, who understands how the immunization occurs, but not for the child, who cannot grasp the process. The test of workability in this case is whether the child ever contracts smallpox.

In applying these tests, at least the first three must be met for us to view the concept, proposition, or outcome as reliable. Failure to pass any of the three (or four) tests discredits the knowledge. Furthermore, meeting these tests does not suggest that the reliable knowledge from the proposition is thereafter unquestionable; that is, passing all four tests is not "proof." It merely indicates that the knowledge is acceptable as working, reliable knowledge. All knowledge is continually under review, scrutiny, and question. Meeting these tests constitutes acceptable evidence that the information, thought, or knowledge is reliable.

While this classification of tests is useful for our thought process about reliability, a central point, or lesson, is that in the final analysis we must consider *all* of the relevant evidence when evaluating reliability of knowledge—disciplinary, subject-matter, and problem-solving. This point is widely recognized by scientists and philosophers (e.g., see Maki, 1996; McCloskey, 1983; Feynman, 1999; Committee on Science, Engineering, and Public Policy, 1995).

ROLE OF PERSONAL OBJECTIVITY

In contemplating matters and questions of science, knowledge and its reliability, and the role and function of research in the overall mixture of these entities, objectivity of the researcher/scientist is the lifeblood of reliable knowledge. Personal objectivity includes avoiding trying to impose our private values and beliefs on others. This does not mean that a completely objective state of mind must always be achieved, but that it be diligently sought. Complete objectivity is probably not a realistic capability of the human condition, but our best effort toward that goal is an aid in seeking reliable public knowledge.

The *subjective*, or subjectivity, is that which is conditioned by personal characteristics of mind; it is associated with beliefs, values, and opinion. Private knowledge is subjective if it is not demonstrable to others. Some things that are normative—what someone thinks should be—are also subjective. Personal judgments or prescriptions that are without logical or empirical support are both normative and subjective. However, all normative matters are not subjective. Knowledge of values is not subjective in a personal sense. The point of the distinction is that we cannot deal with personally subjective matters in science or research because there is no way to derive reliable public knowledge from personal subjective positions by themselves.

The *objective*, or objectivity, is that which is independent of characteristics of mind; it relies on what is demonstrable by observation, measurement, and logic. Things that are positive—identification or specification of what is—are also objective; they are free of personal values and beliefs. Issues, proposals, problems, and questions that are stated in personally objective rather than subjective terms are easier to deal with in the realm of public knowledge. For example, the question of whether the United States should lower import tariffs on a group of goods (a proposition stated subjectively) becomes arguable and is difficult to deal with in a research or scientific context (although often dealt with in a public policy context). Statements about the effects of lower import tariffs on some set of target variables such as domestic consumer and producer prices, national income, balance of trade, and payments are more easily debated as public knowledge. The second set of statements is in judgment-free language that constitutes a valid research question or objective.

Objectivity is widely recognized by researchers and philosophers of science as the cornerstone of science and reliable knowledge. Karl Popper, a well-known founder of the philosophy of pragmatism, emphasized that objectivity in observations, explanations, and criticisms is essential in the pursuit of science (Simkin, 1993). Feynman (1999) believes that the open, questioning, objective state of mind is the most important single element in the pursuit of reliable knowledge. He even argues that it is possible to follow the form and dictates of the scientific method and call it science when it is really "pseudoscience" without objectivity, and there is even a considerable amount of intellectual tyranny in the name of science (pp. 186–188). On the other hand, objective thought can become a habit of thought (p. 248).

What is the realistic capability of researchers and scientists to be personally objective? While that question has no answer in measurable terms, the quantification of objectivity is less important than an awareness of the concept as a goal. French (1971) argues that science has never claimed to be completely objective. We introduce personal subjectivity in our work even by deciding what issues, questions, and problems are more important as objects of study or research. If we remain on guard against the incursion of personal subjectivity into our research work, we produce more reliable research than if we are not diligent. More comprehensive perspectives on this are available in Johnson (1986b, p. 86) and Myrdal (1944, p. 55).

Randall (1974) offers the notion of scientific neutrality as a component of scientific objectivity, and this may be the real value of the goal of objectivity in research. Scientific neutrality embodies value neutrality, avoiding incursion of the scholar's values into his or her scholarly work. However, scientific neutrality cannot ensure neutrality of impact of the results of scholarly work. We must avoid "taking sides" on issues on which there are conflicting or competing interests, but research sometimes results in unequal distribution of benefits. Randall's central point seems to be that scholars and researchers cannot be partisan, though they should recognize that new knowledge has distributional impacts.

As a means of distinguishing between subjective and objective in the research process, the two following distinctions are useful: (1) a proposition or concept is objective if it passes the tests of coherence, correspondence, and clarity; and (2) a researcher or individual is objective if he or she is willing to subject statements to tests of coherence, correspondence, and clarity and abide by the results. Striving to be objective or neutral as researchers may be the most important single goal for us to work toward in the research process.

SCIENTIFIC PREDICTION

Recall that our consideration of concepts that are important for research methodology in economics started with a definition of science as the organized accumulation of systematic, reliable knowledge for the purpose of intelligent explanation or prediction. We then proceeded to consider several aspects of knowledge and establishment of its public reliability. We now will address some conceptual aspects of scientific prediction.

The layman's concept of prediction is typically one of forecasting future events without qualifying conditions on the forecasts. Don't we wish we had the power to perform that kind of feat! Perhaps some people have the ability (gift?) to foretell the future, but it is not within the realm of science or research to achieve such things. As stated earlier, scientific prediction is understood to be conditional because we have no way of knowing with certainty what all of the future conditions that affect an outcome will be. But if we understand how forces interact to produce a result, event, or outcome (i.e., if we can explain it), we have the basis for a conditional prediction. The conditional prediction does not provide as much as we may want, but it may provide the best that is realistically possible. It is also "intellectually honest" in the sense that it makes the conditions of the prediction explicit rather than hides them. For example, an unqualified prediction might be: "GNP next year will increase by 4%." The corresponding conditional prediction might be: "If all the economic sectors perform as we expect (as we assume they will perform), then GNP next year will increase by 4%." Economists are sometimes the focus of ridicule because of their propensity to make the conditions on which their predictions are made explicit (e.g., what we need is more one-armed economists).

The basis of scientific prediction is theory in conjunction with empirical knowledge, including experience. To examine and evaluate this singularly important proposition, we need to define some terms, then examine them and how they are related.

A fact is a verifiable observation.
A theory consists of logical relationships among facts.
A hypothesis is a testable proposition.

The reader is referred to Goode and Hatt (1952, especially pp. 7–17), for extended treatment of these concepts.

Facts constitute a portion of what was previously classed as reliable knowledge. They are observations that we establish through senses and/or measurement, and they can be demonstrated to and verified by other observers. Facts are positive—they establish what is, at least for a specified purpose; they are independent of personal value judgment. Facts, while they constitute reliable knowledge, are not necessarily permanent (Ladd, 1991, p. 8) and are not by themselves useful for prediction. This concept of a fact is different from the popular use of the term, which treats fact as "what is known to be true." This encompasses more than our definition of facts; it encompasses part of what we define as theory. The popular notion of fact is more a vague generalization (i.e., an impression) than a defined term with a specific meaning.

Note that the distinction between facts and values may depend on how a proposition is presented. For example, "Hunger is bad" is a value judgment, but "98% of the U.S. population believes that hunger is bad" may be a fact. Facts may be about knowledge of values. We can make four distinctions about facts and values.[6] There are (1) facts about facts; (2) facts about values; (3) values about facts; and (4) values about values. Items 1 and 2 are proper concerns of science, item 3 is suspect, and item 4 is outside the realm of science. The fact/value dichotomy is often oversimplified and misleading.

For our purposes, we can view facts as having two basic functions (Williams, 1984; Edwards, 1978). One is to initiate theory. Observations of outcomes (facts) in the world we observe gives rise to the need to explain how these facts occur—that is, the need to theorize. Second, facts assist in the reformulation, expansion, and clarification of existing theory. When we observe facts that are not fully consistent with existing theory, the discrepancies must be addressed.

Theory establishes relationships, using facts as building blocks, but is also based on deductive logic—reasoning from given (ideally, but not necessarily) factual premises to their necessary conclusions. The popular understanding of theory is that it is unfounded speculation or a premise without regard to reality or application (e.g., "That may be true in theory, but . . ."). WRONG! Theory is made of carefully thought-out relationships that hold when the specified conditions are met. The only test of a theory is its internal logic.[7] Theory is what gives us predictive/explanatory ability. It permits us to say: "If conditions A, B, C, and D exist, as we anticipate they will, then X and Y should occur." A theory is a model of typical initial conditions that bring relevant phenomena into typical relationships with one another (Simkin, 1993, p. 67). No given theory can possibly fit all circumstances. For a theory to be applicable to a given circumstance—for it to have explanatory capability for a given set of conditions—the premises (assumptions) on which the theory is predicated must be at least a reasonable approximation of the set of conditions to which the theory is being applied. When a specific theory does not fit a given current situation, that does not invalidate the theory; it indicates that it is not applicable for that purpose. Theories may be appropriate at a given time or under given circumstances but not others. Changing conditions can cause theories to become inappropriate or not applicable (Ladd, 1991, p. 8). Also, theory can suggest that a relationship is valid, but not whether it is empirically important (Edwards, 1978, p. 32). Theory cannot provide easy, "magic" answers or prescriptions for intricate and complex problems. Indeed, theories provide us with the framework with which to approach the analysis of these complex problems. Note that theory extends beyond simple causal laws; according to Simkin (1993, p. 65):

> Simple causal laws belong to an early stage of science. As the science develops into a theoretical system it becomes too complex for simple statements. The simple causal laws are limiting cases of scientific explanations that take the general form of probability statements.

Theory thus plays a central role in applied research by providing the conceptual base for specific applications. We may note that the influence also works in the reverse direction. Useful theory draws on reality and application. Theory may be developed in response to recognized real-world problems. If the theorist is not attuned to intended applications, how is he or she to know what to assume and include in the theory?

All theories are abstractions, simplifications, and/or generalizations. They necessarily must be abstractions or simplifications of reality because no reasoning process dealing with relationships can recognize or take account of every variable or condition that could ever affect an outcome. The theory would be so complex that it would be useless. It would apply to only one possible circumstance. Thus, all theory is generalization; it applies to a general class of circumstances. It is this characteristic of theory that makes it useful, because it can be adapted to different specific situations. Thus, theory gives us a framework within which to begin to reason through a given situation or problem to an expected outcome. This kind of application of theory gives rise to hypotheses about expected occurrences.

Heady (1949) identified five functions of economic theory in empirical research. These are to assist in (1) formulating the research problem; (2) formulating hypotheses; (3) designing empirical procedures; (4) assembling appropriate data; and (5) interpreting findings. The functions of theory are not limited to empirical research, however. The functions of theory in research have been addressed by Williams (1984), Edwards (1978, 1990), Ladd (1991), Castle (1989), and Breimyer (1991). Based on these works, the functions of theory in economic research may be listed as:

A. *Orientation*. It provides a framework to examine or delineate a problem or question.

B. *Classification*. It provides a precise means of communication. The set of definitions and specifications embodied in the development of theory provides terms and relationships that have specific, defined meanings that facilitate understanding of complex concepts and phenomena.

C. *Conceptualization*. It gives the means to visualize how something works or to suggest cause and effect relationships that cannot be observed directly. Theory gives us the means to reason through alternate modes of behavior by abstracting to ideal types, giving them rules for behavior, and hypothetically playing them out.

D. *Summarization*. This may be of two types:

1. *Empirical generalizations* such as "Wood floats" or "Prices tend to be inflexible downward."
2. *Generalized relationships*. An example is that the rate of growth in the money supply and the inflation rate are directly related.

E. *Provision of precision* in thought processes. It helps to relate facts and concepts. Data may show correlations, but theory can help sort out cause and effect. Theory assists in thinking clearly and accurately but does not, in itself, provide criteria for solutions to problems or for policy. Problem solving requires other information, which makes it pragmatic and multidisciplinary/interdisciplinary.[8]

F. *Prediction of facts or identification of hypotheses.*

G. *Identification of gaps in our knowledge.* When the body of theory is inadequate to help explain phenomena, there is an inadequacy in our (theoretical) understanding.

The popular notion of theory is often close to our definition of hypothesis. A *hypothesis* is a result or outcome that is not yet evaluated or tested. It is a tentative assertion of a relationship between factors or events that is subject to verification or rejection. It is not necessary, sometimes not even useful, that a hypothesis be "true." A hypothesis must be capable of being accepted or rejected (rejected or not rejected in statistical hypothesis testing) on the basis of the evidence available.

Most researchers, economists included, automatically envision statistical hypothesis testing when the matter of hypotheses arises. In economics, hypothesis testing in econometric research, for example, typically involves testing hypotheses on two levels: (1) testing the statistical significance of individual parameters in the model; and (2) testing the statistical "fit" of the entire model, it being based on expected relationships specified in part or wholly by theoretical considerations. We might refer to statistical hypotheses, and hypothesis testing, as quantitative hypotheses. For enlightening discussions and perspectives on these hypotheses, see Tweeten (1983) and King (1979). Statistical or quantitative hypotheses must have three characteristics to be fully justified: (1) they must have a conceptual basis, that is, be built on theoretical reasoning; (2) they must be sufficiently specific to be rejected or not, based on the data; and (3) there must be data and techniques available to test them.

Another class of hypotheses, less explicitly recognized than quantitative hypotheses but at least as relevant, may be identified as qualitative or conceptual hypotheses (Tweeten, 1983). These hypotheses do not lend themselves to formal, quantitative evaluation. They are used instead as reasonable contentions, assertions, or premises. Although not testable in an empirical sense, they can be agreed upon or not by informed, objective evaluators. These hypotheses are grouped by Tweeten into three categories: maintained, diagnostic, and remedial hypotheses.

Maintained hypotheses are things assumed to be true for purposes of a study being conducted. The assumption that a research objective can be achieved is a maintained hypothesis, and the proposition that, for example, a specified market is structurally competitive is a maintained hypothesis. Acceptance of a maintained hypothesis in the form of general agreement is necessary for the study to proceed and for the results to be accepted.

Diagnostic hypotheses are propositions about the causes of a problem. For example, reasoning that "cheap food" policies of developing countries contributes to low agricultural productivity constitutes a diagnostic hypothesis; alternatively, reasoning that wage controls will lead to labor market inefficiencies in a particular instance is a diagnostic hypothesis. As with maintained hypotheses, acceptance of diagnostic hypotheses as a consensus of analytical reasoning is necessary to establish the premises of a study or the acceptability of results and interpretations. Yet neither is necessarily empirically testable.

Remedial hypotheses are proposed solutions to problems. As Tweeten notes, they are often related to diagnostic hypotheses. They offer prescriptions for potential solutions to problems for which diagnostic hypotheses identify the causes. An example of a remedial hypothesis to the above diagnostic hypothesis example is the prescription: "Removing price controls in country Z would increase food production within the country to a domestic subsistence level." Remedial hypotheses are often conditional ("The goal of X can be achieved by action Y"), and they frequently appear in the implications or conclusions of research. These hypotheses, as with the other two, are typically not empirically testable but are accepted or rejected by weight of evidence by reason.

SUMMARY

This chapter provides definitions and concepts of science, knowledge, theory, and personal objectivity and examines their relationships to research methodology in economics. Science—the organized accumulation of systematic, reliable knowledge—provides a knowledge base for research. Research, in turn, adds to the body of science and is the vehicle for validating scientific knowledge. Economics is presented, along with other sciences, as being scientific in its discipline knowledge and research methodology. Extending the science to multifaceted contemporary societal problems and issues often requires multidisciplinary/interdisciplinary research, judgments and decisions.

Knowledge is classified in two ways: (1) positivistic and normativistic (prescriptive or knowledge of values); and (2) private and public. Both are useful in evaluating and understanding knowledge generated through research, although only unconditionally prescriptive and private (personally subjective) kinds of knowledge are not valid within the context of research activities. The basic avenues for obtaining knowledge—the senses, experience, intuition, revelation, measurement, and reasoning—are examined in relation to *reliable* knowledge. Tests for reliability of knowledge are the tests of correspondence, coherence, and clarity, and some wish to add the pragmatic test of workability. In a more general, strictly logical vein, eight logical fallacies are also presented.

Personal objectivity as a state of mind of researchers is a most important goal in economic research. The role of objectivity is to avoid bias in the production of knowledge, not to deny the place of values, beliefs, and opinions in one's activities.

Facts, theories, and hypotheses are defined and the roles of each in science and scientific prediction are addressed. The relationships of facts, theories, and hypotheses to each other is covered, their functions identified, and popular misconceptions about them are discussed. Various classes of hypotheses—quantitative and qualitative—are also introduced.

RECOMMENDED READING

Johnson. *Research Methodology for Economists*, pp. 16–20, 43–49.
Larrabee. *Reliable Knowledge*, pp. 1–24, 104–106.
Randall. "Information, Power, and Academic Responsibility."

SUGGESTED EXERCISES

1. Describe a specific instance of an application of economics in which it represents "science," then describe a specific instance of an application of economics as "art." Do the same thing for a physical science (biology, chemistry physics, geology, etc.).
2. Does one's political philosophy classify as public or private knowledge? Explain.
3. Describe an example of each of the eight logical fallacies based on your own experience.
4. Select a journal article from any recent issue of an economics journal. Identify the first five subjective statements in the article and (a) explain why/how each statement is subjective; and (b) change it to an objective statement.
5. Describe an example, from your own experience, of a situation in which different people have (or would have) taken the same premise or proposition and interpreted it alternatively as comprising a fact, a theory, and a hypothesis.
6. Provide a written description of each of the three types of qualitative hypotheses identified in this chapter.

NOTES

1. This notion may be related to Johnson's (1986b) thoughts on interpersonal validity (pp. 241, 242), but Larrabee's discussion is much more extensive.

2. Johnson (1986b, pp. 228–229) argues that all descriptive knowledge rests on "faith" because it cannot be proven.

3. Unless efficiency is clearly defined as to its criteria, the statement remains a normatively stated proposition because "efficiency" presumes certain objectives. For elaboration of this point, see Ladd (1983, especially pp. 1–3).

4. Some maintain that all logic is necessarily deductive, and this precept is accurate by most historical standards. However, induction such as that embodied in statistical hypothesis testing is inherently as logical as deductive reasoning. It may be noted that the theoretical basis of statistical tests relies on deduction (reasoning from premises or assumptions to conclusions). Thus, on careful examination it may not be clear whether the basis of the reasoning is inductive or deductive. Nevertheless, inductive logic is used here to be synonymous with inductive reasoning.

5. The author is indebted to Josef Broder for this example.

6. This perspective was provided by Josef Broder.

7. Some maintain that the conclusive test of a theory is its ability to predict. However, this confuses the logic of a theory with applicability to a particular situation. That is, when a theory does not explain a given situation it does not apply to that situation, but the theory may explain other situations. The theory of perfectly competitive market behavior is a useful example. Just because that theory does not explain pricing behavior in the U.S. automobile industry does not invalidate the theory of perfectly competitive markets. The premises of the theory (the conditions for perfect competition) are not consistent with the conditions in the U.S. auto industry. Thus, that theory is not applicable to that particular market situation. But the assumptions of the theory are close enough to the conditions existing in the wheat-producing sector of the U.S. economy that competitive market theory is useful in explaining behavior in that industry. Eichner (1986, p. 5) notes that economists reject the view that theories must be empirically confirmed (a view held strongly by positivists). Within economics, empirical confirmation is valued more highly by applied than by disciplinary economists. Simkin (1993, p. 16) argues that demarcation of theories according to testability is not the same thing as demarcation according to intellectual worth.

8. The distinction between multidisciplinary and interdisciplinary research is as follows. Multidisciplinary research uses concepts and/or tools from two or more disciplines; the different disciplines contribute to the research process, usually with researchers from the separate disciplines working together in tandem. Problem-solving research abounds with examples of multidisciplinary research. Interdisciplinary research goes further to actually integrate concepts from different disciplines in the research process. Subject-matter research offers many interdisciplinary examples. Consider that environmental or natural-resource economics often integrates biological concepts with economic concepts in research in that subject-matter field.